

2025

(Session : 2024-27)

(Paper ID : 12006)

Time : 3 hours

Full Marks : 80

Note: Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

Answer any five questions.

1. Define the following terms :

- a. Venn Diagram
- b. Proper Subset
- c. Complement of a Set
- d. Singleton Set

2. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6\}$ and $C = \{x : x \in \mathbb{Z} \text{ is even}\}$. Find:

- a. $A - B$
- b. $B \cap A$
- c. $C - B$
- d. $(A \cap B) \cup C$

3. What is function ? Define it. Explain various types of functions with suitable example.

4. Define Group with an example.

5. List all partition of sets :

- a. (a) $A = \{a, b, c\}$
- b. (b) $A = \{1, 2, 3, 4\}$

6. In a class of 25 students 12 have taken Mathematics, 8 have taken Mathematics but not Biology. Find the number of students who have taken Mathematics and Biology and those who have taken Biology but not Mathematics.

7. A tree has $2n$ vertices of degree 1, $3n$ vertices of degree 2 and n vertices of degree 3. Determine the number of vertices and edge in the tree.

8. Draw the Hasse diagram for the poset $(P(S) \subseteq)$ where $P(S)$ is the power set on $S = \{a, b, c\}$.

9. If $A = \{1, 3, 5\}$, let R be a relation such that $XY : \text{if } Y = X + 2$ and S be the relation such that $XS : \text{if } X < Y$ then find the following:

- a. $R \circ S$
- b. $S \circ R$
- c. $R \circ R$
- d. $S \circ S$

with the help of diagram.

10. Write short notes on the following :

- a. Monoid
- b. Partition of set
- c. Rings
- d. Binary Operation

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